Non-Equilibrium Properties Reveal the Limits of Specialist and Foundation Machine-Learned Force Fields

Yi Cao[1], Peter Mastracco[1], Paulette Clancy[1*]

[1]Department of Chemical and Biomolecular Engineering, Johns Hopkins University, 3400 N. Charles Street, Baltimore, 21218, Maryland, USA.

*Corresponding author(s). E-mail(s): pclancy3@jhu.edu;
Contributing authors: ycao73@jh.edu; pmastra2@jh.edu;

Abstract

Machine-learned force fields (MLFFs), particularly those arising from pre-trained foundation models, offer an enticing compromise between *ab initio* accuracy and the larger scope possible with semi-empirical force field-derived molecular dynamics with respect to length and time scales. Despite their promise, a fundamental question remains: To what extent should one rely on specialist models, fine-tuned foundation models, or a hybrid approach? Trade-offs in data efficiency, robustness, and non-equilibrium fidelity remain poorly understood, as does understanding how different training paradigms encode physical information. To assist with this decision, we introduce a unified benchmarking framework based on non-equilibrium configurational probes that span interpolation and extrapolation regimes. Migration pathways computed via Nudged Elastic Band calculations serve as the primary diagnostic probe, complemented by equilibrium and mechanical tests. Using Cr-doped Sb_2Te_3 as a representative 2D material for which non-equilibrium properties are critically important, we evaluate multiple training strategies within the MACE architecture. While all models reproduce equilibrium structures, they diverge sharply in their ability to adequately capture non-equilibrium processes. Fine-tuned MACE models improve selected kinetic probes but can degrade long-range mechanical response. "Zero-shot" MACE foundation models exhibit overly smoothed potentials and specialist models can fail for high-energy transitions. Latent-space and surrogate-error diagnostics are consistent with distinct model representations across training paradigms, supporting the conclusion that equilibrium metrics alone are insufficient.

This work demonstrates migration-based non-equilibrium probes as a data-efficient candidate framework for MLFF evaluation, developed and tested on a representative material system for which such probes are important, and provides practical guidance for robust MLFF development.

Keywords: Machine-learned force fields (MLFFs), foundation models, fine-tuning, defect migration pathways, 2D materials

1 Introduction

Machine-Learned Force Fields (MLFFs) [3**? ? ? ? ? ? ? ? ? ? ? ? ? ?**] have emerged as a transformative solution to the dilemma of accuracy-versus-cost in molecular simulation. The development of large-scale, pre-trained so-called "foundation models," such as CHGNet, [**?**] MEGNet, [**?**] M3GNet, [**?**] MACE-MP, [3] MatterSim, [**?**] and UMA, [**?**] marks a paradigm shift to try to resolve this dilemma. [**?**]. However, this shift presents researchers with a critical question: For a specific system, is it more effective to invest significant resources to build a robust *specialist* model from scratch, [**? ? ?**] or to adapt a *generalist* foundation model through fine-tuning? While generic benchmarks like Matbench, [**?**] JARVIS-Leaderboard, [**?**] and LeMat-GenBench, [**?**] validate their broad utility to predict equilibrium properties for which they were trained, their reliability for specific, high-fidelity applications remains an open question. [**? ? ?**] A valid concern is that these models fail to capture potentially more nuanced interactions that govern critical processes that shape the materials and their processing. [**? ? ? ?**] Fine-tuning is a promising solution, but using naïve strategies can lead to *"catastrophic forgetting"*, where general knowledge about the system is erased. Key questions of data efficiency, [**? ?**] stability, [**? ?**] and transferability [**? ?**] for non-equilibrium processes remain largely unsolved. [**? ? ?**]

To build a rigorous benchmark that addresses the dilemma facing the specialist versus the generalist, we must first define what distribution and what observables are being measured. Conventional MLFF validation typically estimates pointwise energy and force errors on near-equilibrium configurations. Such tests are necessary, but they do not, by themselves, measure whether the learned potential-energy surface is reliable along the low-probability configurations that control kinetic and mechanical processes.

To test this hypothesis, we formulate migration and collective-displacement pathways as task-conditioned [Yi- what is meant by a "task-conditioned probe"? The wording is rather non-specific] probes of the potential energy surface. In this view, a pathway is not merely an application-specific calculation; it is a structured evaluation distribution that explicitly samples configurations between metastable states, including transition-state and strained-lattice regions that may well be rare or absent in equilibrium datasets. [Yi- That sentence is far too long and it is not direct enough for me to understand what you intend to say. Let's discuss.] The mathematical framework used to separate equilibrium pointwise accuracy, fixed-path energetic fidelity, and self-relaxed path stability is defined in Sec. **??**.

To demonstrate and validate this benchmarking framework, we employ doped two-dimensional (2D) van der Waals (vdW) materials as an ideal test bed. The discovery and design of novel 2D vdW materials [**? ? ? ? ?**] continues to drive innovation in spintronics, [**?**] quantum devices, [**? ? ?**] and energy technologies. [**? ?**] A particularly powerful strategy for tuning the properties of these layered materials involves doping, the insertion of "foreign" atoms between vdW layers. [**? ?**] This approach has enabled remarkable advances in materials design, ranging from lithium-ion batteries [**? ?**] to the engineering of magnetic semiconductors [**?**] and quantum anomalous Hall systems. [**?**] Crucially, their layered structure offers distinct environments for atomic motion, making them an excellent probe for testing MLFF capabilities across different chemical contexts within a single system.

Among 2D materials, topological insulators such as antimony telluride (Sb_2Te_3) [**? ?**] represent an especially promising class for doping studies. The doping of transition metals has proven to be effective across multiple topological systems: Iron in Bi_2Se_3 creates ferromagnetic order [**? ? ?**]. Manganese in Bi_2Te_3 enables tunable magnetic properties. [**? ?**] And vanadium in Sb_2Te_3 modifies its electronic structure. [**? ?**] The doping of chromium (Cr) into Sb_2Te_3, the focus of this work, offers a compelling route to engineer its magnetic and topological properties for spintronic applications. [**? ? ?**] Predicting the stability and dynamics of these complex systems is essential for guiding experimental synthesis, a task for which large-scale Molecular Dynamics (MD) simulations are extremely powerful [**? ?**] provided that a suitably representative force field is available. However, the atomic-scale processes governing functionality, including dopant migration and thermal stability, occur on time- and length- scales far beyond the reach of first-principles methods like density functional theory (DFT). [**? ? ? ?**]

This work addresses these gaps by presenting a systematic benchmark of bespoke training and fine-tuning strategies for the case study of Cr-doped Sb_2Te_3 using the MACE architecture (Fig. **??**). By combining latent-space analysis with migration-based probes, we move beyond traditional accuracy metrics to evaluate the dynamic stability and kinetic predictions of each model. Our study aims to serve as both a practical guide for selecting MLFF training strategies and a diagnostic framework for designing more data-efficient learning loops. This manuscript extends our earlier report [**?**] by adding a unified fixed-path and self-consistent NEB analysis, interlayer sliding tests, representation diagnostics, and a data provenance workflow. It therefore treats migration pathways not only as an application result, but also as a structured benchmark distribution for testing model behavior under task-relevant distribution shift. [Yi- I think this sentence was repeated earlier. It needs rewording to be easier to understand.] This revised framing also positions the work relative to recent studies of MACE fine-tuning for ion diffusion, foundation-MLIP migration-barrier evaluation, and systematic softening of universal MLIP potential-energy surfaces [PLACEHOLDER: final nearest-neighbor literature citation-key audit].[Yi- needs a reference and a bit more explanation: How is this work similar/different to other studies?]

To this end, we address the following key questions:

1. How do bespoke MLFFs, trained from scratch, compare against large pre-trained foundation models in terms of accuracy, stability, and data efficiency for molecular

simulation? What are the performance and stability trade-offs of different fine-tuning strategies?

2. Can non-equilibrium probes generated from methods, such as NEB, provide an efficient benchmark for specialist versus generalist models, and what additional cross-system evidence is needed before such probes can be considered generalizable across materials and architectures?
3. Can latent-space analysis reveal failure signatures to guide data acquisition?

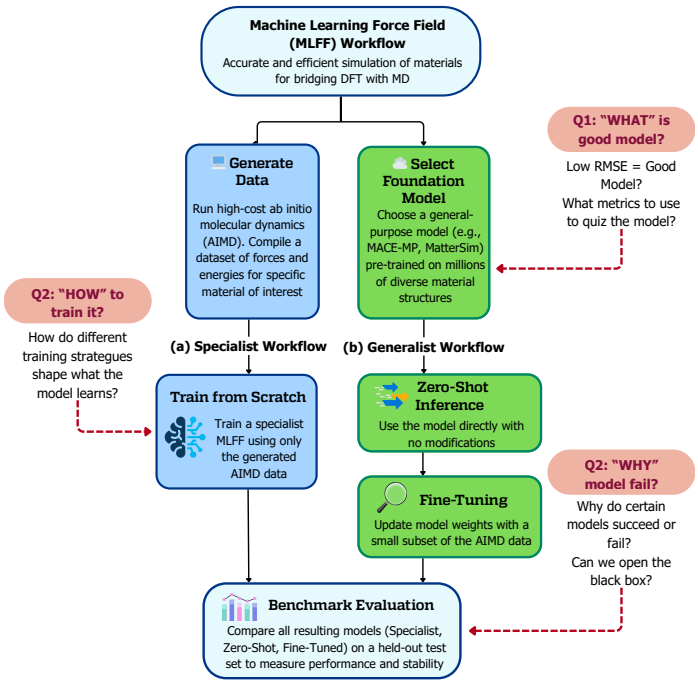


Fig. 1: **Conceptual Workflow Diagram** illustrating two competing workflows: (a) training a specialist MLFF from scratch and (b) fine-tuning a generalist foundation model. The diagram highlights the benchmarking questions addressed in this paper.

2 Methodology

We employed a systematic approach to benchmark various Machine-Learned Force Fields (MLFFs) for Cr-doped Sb_2Te_3[1], comparing specialist models trained from scratch against fine-tuned foundation models.

[1]This material system serves as an ideal testbed because its anisotropic van der Waals structure provides distinct migration environments: in-gap diffusion within quintuple layers (representing in-distribution scenarios) and interlayer migration across vdW gaps (probing out-of-distribution regimes). This structural duality enables efficient sampling of diverse configurations—from equilibrium states to high-energy transition states—while addressing technologically relevant processes where both thermodynamic stability and kinetic properties are critical for doping engineering.

2.1 Data-Generation and Model-Training

Reference data were generated using density functional theory (DFT) calculations adopting the well-used PBE functional [**?**] and comprising roughly 20,000 configurations of a $4 \times 2 \times 1$ system of the Sb_2Te_3 crystal [**?**]. We use a 1 fs timestep for a series of *ab initio* molecular dynamics (AIMD) simulations at three temperatures (300 K, 600 K, 1200 K). We employed the MACE architecture for all models, which is a message-passing neural network for the representation of atomic interactions. [3]

This paper evaluates four distinct training strategies to benchmark their performance, named as follows:

1. **Scratch**: MACE model trained exclusively on our AIMD dataset
2. **Foundation**: Pre-trained MACE-MP model used without fine-tuning
3. **FT-600 K**: Foundation model fine-tuned on 5% of data from 600 K trajectories
4. **FT-MultiT**: Foundation model fine-tuned on multi-temperature data from the three temperatures studied

Revision 1 treats frame-level train/test splitting as insufficient for benchmark claims involving migration pathways. The revised data workflow assigns each entire source trajectory to a single partition (train, validation, or test) rather than shuffling individual frames, so that no trajectory is split across partitions. Under this grouped protocol (1,786/311/267 train/validation/test frames), the nearest-neighbor SOAP cosine distance between any in-gap NEB image and the grouped training set is 5.4×10^{-4}, and the closest comparable Cartesian RMSD is 2.06 Å; both are far outside the exact/near-duplicate thresholds used for this audit (SOAP $\leq 10^{-4}$, RMSD ≤ 0.05 Å), so no NEB-image/training-frame overlap was detected under the declared metrics. The Scratch-5% training subset (89 frames) gives the same result (SOAP 6.4×10^{-4}, RMSD 2.06 Å). Grouped-split retraining across seeds 123/234/345 is complete for FT-600K fine-tuned from the OMAT-FT checkpoint, Scratch, and Scratch-5%, together with a single-seed MACE-MPA-0 fine-tuned comparator (excluded from 3-seed uncertainty estimates); the resulting RMSE and in-gap barrier statistics are reported below and in Table S1.

The reported fine-tuned models are documented as being trained on AIMD frames with real DFT energies and forces, not on NEB-derived exports lacking real force labels. The revised data manifest records source trajectory, split assignment, energy/force key mapping, and checksums for model-training files. The provenance audit covers 9,163 frames across the historical and grouped exports; the only nine exact zero-force frames are explicitly labelled one-atom Cr/Sb/Te isolated-atom E_0 references, with zero unclassified and zero NEB-derived zero-force frames. The complete manifest and audit code are in release commit `f5c6d19`.

2.2 Evaluation Framework

Models were assessed through a hierarchy of measurement operators that separate near-equilibrium pointwise accuracy from task-conditioned process fidelity. Static energy and force errors, molecular dynamics observables, fixed-path NEB scoring, self-relaxed NEB stability, interlayer sliding, and representation diagnostics were evaluated as

distinct probes of the same candidate force fields. Detailed computational parameters are provided in the Supplementary Information.

2.3 Formal Benchmark Definition

The central object of this benchmark is not a single scalar RMSE, but a family of task-conditioned measurement operators applied to the same candidate force field. Let a configuration be denoted by

$$x = (\mathbf{R}, \mathbf{Z}, \mathbf{h}) \in \mathcal{X},$$

where $\mathbf{R}$ contains atomic coordinates, $\mathbf{Z}$ atomic identities, and $\mathbf{h}$ the simulation cell. A reference electronic-structure model defines an energy $E_{\mathrm{D}}(x)$ and force field $\mathbf{F}_{\mathrm{D}}(x) = -\nabla_{\mathbf{R}} E_{\mathrm{D}}(x)$. A candidate MLFF M defines $E_M(x)$ and $\mathbf{F}_M(x) = -\nabla_{\mathbf{R}} E_M(x)$.

Definition 1 (Equilibrium and task-conditioned evaluation distributions) The conventional equilibrium test distribution is

$$\mu_{\mathrm{eq}}(x) \propto \exp[-\beta E_{\mathrm{D}}(x)]$$

restricted in practice to relaxed structures, short AIMD trajectories, or random splits of near-equilibrium DFT configurations. Its standard pointwise error is

$$\mathcal{E}_{\mathrm{pt}}(M; \mu_{\mathrm{eq}}) = \left(\mathbb{E}_{x \sim \mu_{\mathrm{eq}}} \left[w_E |E_M(x) - E_{\mathrm{D}}(x)|^2 + w_F \|\mathbf{F}_M(x) - \mathbf{F}_{\mathrm{D}}(x)\|^2 \right] \right)^{1/2}.$$

For a physical process τ, such as dopant migration or interlayer sliding, the benchmark distribution is a process-conditioned distribution μ_τ supported on a structured set of configurations generated by a path operator. This distribution may place substantial probability mass on strained, high-energy, low-Boltzmann-weight configurations that are rare under μ_{eq} but control kinetics and mechanical stability.

Definition **??** makes explicit why equilibrium RMSE can be accurate and still incomplete: it measures the average pointwise error under μ_{eq}, whereas barrier prediction, diffusion, and sliding depend on E_M and ∇E_M over regions sampled by μ_τ. The two tests therefore answer different mathematical questions.

2.3.1 Pointwise Probes and Process Probes

A pointwise probe evaluates a model on isolated configurations,

$$\mathcal{P}_{\mathrm{pt}}(M; \mathcal{S}) = \{E_M(x), \mathbf{F}_M(x) : x \in \mathcal{S}\},$$

where $\mathcal{S}$ is a fixed set of DFT-labeled structures. This includes static energy/force RMSE and RDF-preserving near-equilibrium MD checks. A process probe evaluates a functional of a correlated configuration sequence,

$$\Gamma = (x_0, x_1, \ldots, x_N),$$

or of a trajectory distribution generated by M. Examples include a migration barrier, endpoint asymmetry, NEB convergence, diffusion coefficient, thermal conductivity, and sliding periodicity. The benchmark therefore measures both local correctness,

$$E_M(x_i) \approx E_{\mathrm{D}}(x_i), \qquad \mathbf{F}_M(x_i) \approx \mathbf{F}_{\mathrm{D}}(x_i),$$

and functional correctness,

$$\Phi_M(\Gamma) \approx \Phi_{\mathrm{D}}(\Gamma),$$

where Φ is a task-level observable rather than a pointwise label.

2.3.2 Migration-Path Selection

Let $\mathcal{A} = \{a_1, \ldots, a_K\}$ be metastable dopant sites obtained from relaxed DFT structures or thermally sampled AIMD motifs. A migration task is an ordered pair

$$\tau = (a_s, a_t, c),$$

where a_s and a_t are initial and final sites and c records the chemical and geometric context, such as in-gap motion, penetration into a quintuple layer, or interlayer sliding. The selected task set

$$\mathcal{T} = \mathcal{T}_{\mathrm{ID}} \cup \mathcal{T}_{\mathrm{OOD}} \cup \mathcal{T}_{\mathrm{coll}}$$

is designed to include: (i) interpolation-like paths whose local environments overlap with AIMD training data, (ii) extrapolation-like paths that pass through strained or high-energy transition states, and (iii) collective displacements where local environments change weakly but long-range registry or periodicity matters.

For each τ, a path is initialized by interpolation or chemically constrained image construction,

$$\Gamma_\tau^0 = (x_0, \ldots, x_N), \qquad x_0 = a_s, \quad x_N = a_t,$$

and then optimized by a reference or model-specific NEB operator. A path is retained as a benchmark probe only if its endpoints, chemical identity, cell convention, and image ordering are auditable from the raw inputs and outputs.

2.3.3 Fixed-Path and Self-Relaxed NEB Operators

The apparent discrepancy between a fixed-path barrier in Fig. **??** and a self-consistent MLFF-NEB barrier in Fig. S3 is resolved by defining two different operators.

Reference DFT path.

For a task τ, the DFT NEB operator returns a converged reference path

$$\Gamma_{\mathrm{D},\tau}^* = \mathrm{NEB}_{\mathrm{D}}(x_0, x_N) = (x_{0,\tau}^{\mathrm{D},*}, \ldots, x_{N,\tau}^{\mathrm{D},*}).$$

The DFT barrier is

$$\Delta E_{\mathrm{D}}^{\ddagger}(\tau) = \max_{0 \le i \le N} E_{\mathrm{D}}(x_{i,\tau}^{\mathrm{D},*}) - E_{\mathrm{D}}(x_{0,\tau}^{\mathrm{D},*}).$$

Fixed-path scoring.

The fixed-path MLFF operator evaluates model energies on the same DFT-relaxed images:

$$\Delta E^{\ddagger}_{\text{fixed}}(M,\tau) = \max_{0\le i\le N} E_M(x^{\mathrm{D},*}_{i,\tau}) - E_M(x^{\mathrm{D},*}_{0,\tau}).$$

It measures whether M assigns the correct potential-energy profile to an externally supplied DFT path. The model does not choose or relax the path.

Self-relaxed MLFF NEB.

The self-relaxed operator lets the MLFF define the forces that optimize the band:

$$\Gamma^{*}_{M,\tau} = \mathrm{NEB}_M(x_0, x_N) = (x^{M,*}_{0,\tau}, \ldots, x^{M,*}_{N,\tau}),$$

followed by

$$\Delta E^{\ddagger}_{\text{self}}(M,\tau) = \max_{0\le i\le N} E_M(x^{M,*}_{i,\tau}) - E_M(x^{M,*}_{0,\tau}).$$

This operator measures optimization stability and qualitative path robustness on the MLFF's own PES. Because $\Gamma^{*}_{M,\tau}$ and $\Gamma^{*}_{\mathrm{D},\tau}$ need not be the same curve in configuration space, $\Delta E^{\ddagger}_{\text{fixed}}$ and $\Delta E^{\ddagger}_{\text{self}}$ are not expected to agree. A low self-relaxed MLFF barrier is physically validated only after the path is checked against DFT, for example by DFT single-point energies on $\Gamma^{*}_{M,\tau}$ or by restarting a DFT NEB from the MLFF path.

2.3.4 Error Decomposition

The fixed-path barrier error is

$$\epsilon_{\text{fixed}}(M,\tau) = \Delta E^{\ddagger}_{\text{fixed}}(M,\tau) - \Delta E^{\ddagger}_{\mathrm{D}}(\tau).$$

Define the local energy residual and the model/reference maximum-image indices as

$$\delta_M(x) = E_M(x) - E_{\mathrm{D}}(x),$$
$$i^{*}_{M} = \arg\max_i E_M(x^{\mathrm{D},*}_{i,\tau}), \qquad i^{*}_{D} = \arg\max_i E_{\mathrm{D}}(x^{\mathrm{D},*}_{i,\tau}).$$

The error can then be separated into endpoint and saddle-region terms:

$$\epsilon_{\text{fixed}} = \underbrace{\delta_M(x^{\mathrm{D},*}_{i^{*}_{D}}) - \delta_M(x^{\mathrm{D},*}_{0})}_{\text{endpoint-to-saddle energy calibration}} + \underbrace{E_M(x^{\mathrm{D},*}_{i^{*}_{M}}) - E_M(x^{\mathrm{D},*}_{i^{*}_{D}})}_{\text{maximum-image localization on the fixed path}}.$$

Thus a model can fail by shifting the endpoint energy, miscalibrating the saddle energy, or moving the apparent maximum along the path. For a self-relaxed NEB, an additional

geometric term appears because the model samples a different path:

$$\epsilon_{\text{self}\to\text{D}}(M,\tau) = \left[\max_{x_i \in \Gamma^{\star}_{M,\tau}} E_{\text{D}}(x_i) - E_{\text{D}}(x_0)\right] - \Delta E^{\ddagger}_{\text{D}}(\tau),$$

which measures whether the MLFF-discovered path is also plausible on the DFT PES. In practice this term requires DFT single-point calculations or DFT NEB refinement on the MLFF-relaxed path.

2.3.5 Robustness and Generalization Claims

The benchmark supports claims at three levels, with different evidence requirements.

1. *Pointwise accuracy*: $\mathcal{E}_{\text{pt}}(M;\mu_{\text{eq}})$ is small, as measured by energy/force RMSE, RDF agreement, and short-time stability.
2. *Process fidelity*: $|\Phi_M(\tau) - \Phi_{\text{D}}(\tau)|$ is small for a specified task τ, as measured by barriers, transport observables, sliding profiles, or DFT-path scoring.
3. *Robust generalization*: Φ_M remains accurate over a documented task family $\mathcal{T}$, with multi-path, multi-seed, OOD, and convergence checks.

The present work claims Level 2 evidence for selected Cr-doped Sb_2Te_3 tasks and proposes Level 3 as a benchmark design principle requiring broader validation across materials, migration chemistries, architectures, and random seeds. This distinction prevents a successful result on one path from being over-interpreted as universal transferability.

2.3.6 Benchmark Algorithm

Algorithm 1 Migration-centered MLFF benchmark

Define candidate models $\mathcal{M} = \{M_1, \ldots, M_L\}$ and reference method D.
Construct or audit equilibrium data $\mathcal{S}_{\mathrm{eq}}$ from DFT relaxations and AIMD.
Train or adapt models using documented training splits, seeds, and code hashes.
Measure pointwise errors $\mathcal{E}_{\mathrm{pt}}(M; \mathcal{S}_{\mathrm{eq}})$ and near-equilibrium observables.
Select migration and collective-displacement tasks $\mathcal{T} = \mathcal{T}_{\mathrm{ID}} \cup \mathcal{T}_{\mathrm{OOD}} \cup \mathcal{T}_{\mathrm{coll}}$.
for each task $\tau \in \mathcal{T}$ **do**
    Generate endpoint-verified initial image sequence Γ_τ^0.
    Compute or restart $\Gamma_{\mathrm{D},\tau}^* = \mathrm{NEB}_{\mathrm{D}}(x_0, x_N)$ until convergence.
    Score each M on the fixed DFT path to obtain $\Delta E_{\mathrm{fixed}}^{\ddagger}(M, \tau)$.
    Optionally run $\Gamma_{M,\tau}^* = \mathrm{NEB}_M(x_0, x_N)$ to assess self-relaxed stability.
    If $\Gamma_{M,\tau}^*$ differs from $\Gamma_{\mathrm{D},\tau}^*$, validate the MLFF path with DFT single points or a DFT restart.
    Record barrier errors, endpoint errors, maximum-force histories, convergence flags, RMSD/SOAP overlap, and checksums.
end for
Report model rankings with uncertainty over seeds, paths, and convergence status.
Interpret latent-space or SHAP diagnostics as supporting evidence, not as a replacement for task-level validation.

2.3.7 Applicability Assumptions and Acceptance Criteria

The framework is applicable when the process of interest can be represented by auditable endpoint states and a physically meaningful path or trajectory functional. It is especially appropriate for rare events, migration, sliding, stacking rearrangements, and other processes where a small set of high-leverage configurations controls the downstream observable. It is less complete when the dominant failure mode involves electronic excited states, bond-breaking chemistry absent from the reference functional, long-range electrostatics beyond the model architecture, or finite-size effects not represented in the chosen cell.

A benchmark task is accepted into the final manuscript only if the following checks are satisfied:

1. DFT endpoints and NEB images are reconstructable from raw inputs and outputs.
2. The DFT reference barrier is convergence-verified, or explicitly labeled as a current candidate value.
3. Fixed-path and self-relaxed NEB quantities are reported as distinct operators.
4. Model comparison includes seed sensitivity where training stochasticity can affect rankings.
5. Training/test leakage is checked with grouped splits or path-aware proximity metrics.
6. Path overlap is quantified with geometric RMSD and/or descriptor-space similarity.

7. Any MLFF-discovered low-barrier path is tested on the DFT PES before being interpreted physically.
8. Scripts, model identifiers, input files, convergence flags, and output checksums are preserved in the release manifest.

2.3.8 Physics and Data-Evidence Chain

The physical basis of the benchmark is transition-state theory: kinetic rates depend exponentially on activation free energies, so small barrier errors can lead to large rate errors,

$$k \propto \exp[-\Delta G^{\ddagger}/(k_{\mathrm{B}}T)].$$

Consequently, a model that has low average force RMSE near equilibrium can still be unreliable if it misrepresents the PES along rare-event pathways. The evidence chain used here is:

$$\begin{aligned}\text{DFT/AIMD data} &\rightarrow \text{trained MLFFs} \rightarrow \text{equilibrium validation}\\ &\rightarrow \text{path/process probes} \rightarrow \text{DFT ground-truth comparison}\\ &\rightarrow \text{representation diagnostics} \rightarrow \text{data-acquisition guidance.}\end{aligned}$$

This chain closes the benchmark loop: physical theory identifies the observables that matter, DFT supplies the reference PES, MLFFs are evaluated through measurement operators matched to downstream processes, and diagnostic analyses indicate which missing configurations should be acquired next.

2.4 SHAP Analysis

We employed a model-agnostic SHAP workflow to identify SOAP descriptors associated with MACE energy-error behavior. Local atomic environments were encoded using SOAP descriptors (5.0 Å cutoff, $n_{\max} = 4$, $l_{\max} = 4$), capturing species-resolved radial and angular interactions in Cr-doped Sb_2Te_3. For each MACE variant, gradient-boosted regressors (200 trees, learning rate 0.05, max depth 8) were trained to predict the absolute total-energy error on the audited 150-frame trajectory. TreeSHAP values were then computed for the fitted surrogate models, and surrogate fidelity was assessed with 5-fold cross-validation rather than in-sample R^2. We compared the top 50 influential SOAP features across models to identify shared versus model-specific sensitivities in the surrogate error model.

3 Results and Discussions

3.1 Equilibrium Properties: Foundation for Comparison

As a baseline, we first evaluated all four candidate models on the near-equilibrium DFT data used in conventional MLFF validation. In the notation of Sec. **??**, this corresponds to estimating a pointwise error under μ_{eq}, the distribution of relaxed and thermally sampled configurations near local minima. All models satisfy typical energy and force RMSE thresholds (Table S1), and the corresponding parity plots show visually similar agreement with DFT across energies and forces (Fig. S1).

The convergence of these pointwise metrics does not imply equivalent behavior under the task-conditioned distributions μ_τ associated with migration, sliding, or rare-event kinetics. We therefore treat equilibrium accuracy as a necessary baseline rather than a sufficient validation criterion, [Yi- what does that mean?] and next evaluate each model using process probes whose observables depend on the shape of the potential-energy surface along physically motivated paths. [Yi- such as?]

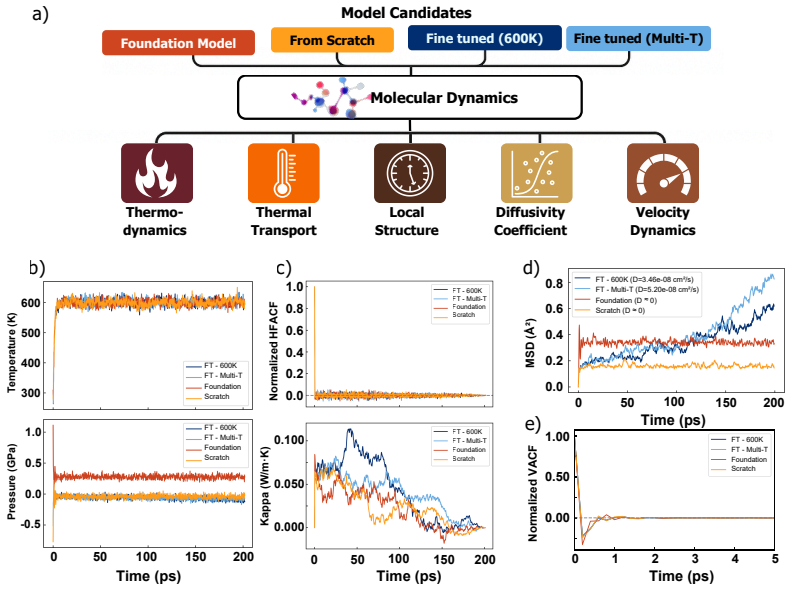


Fig. 2: **Comprehensive Benchmarking of MLFF Performance in Molecular Dynamics Simulations.** (a) A schematic of the unified evaluation protocol. Four candidate models (a zero-shot foundation model, a bespoke model trained from scratch, and two fine-tuned variants) were used to drive a 200 ps MD simulation. The resulting trajectories are then subjected to a uniform set of post-processing analyses to evaluate key physical properties. (b) Thermodynamic stability, demonstrated by the evolution of temperature and pressure over the simulation, remain acceptably stable around their target values for all models. (c) Thermal transport properties, showing the Heat Flux Autocorrelation Function (HFACF) and its running integral to compute thermal conductivity (κ). (d) The Mean Squared Displacement (MSD), used to assess atomic mobility and calculate the diffusion coefficient. (e) The Velocity Autocorrelation Function (VACF), which describes the system's underlying dynamics.

All MACE models (Foundation, Scratch, FT-600K, FT-Multi_T) successfully reproduce the equilibrium structure of Cr-doped Sb_2Te_3, with RDF analysis showing excellent agreement with the AIMD reference ("ground truth") across all atomic pair correlations (Fig. S2). This Figure includes a structural schematic of the local environment around Cr and full radial distribution functions (RDFs) for all models, confirming

that each training strategy preserves the characteristic nearest-neighbor and second-shell coordination features at 600 K. While all models maintain stable thermodynamic ensembles at 600 K (Fig. **??**c), the zero-shot foundation model shows a persistent pressure offset, likely originating from its exposure to predominantly 0 K equilibrium structures rather than configurations extant at finite temperatures.

Despite similar performance on local structural (RDF) and short-time dynamic (VACF) properties (Fig. **??**e), the models diverge in their preliminary predictions of longer timescale transport phenomena. The original FT–600K transport calculation used a non-comparable trajectory protocol [Yi- how do you know that about the trajectory?] and is therefore quarantined [Yi- what does "quarantined" mean in this context?] rather than used for final model ranking. Because diffusion and thermal transport coefficients are sensitive to sampling length and trajectory health, the final analysis will replace these values with the corrected matched long-run MD workflow [PLACEHOLDER: corrected 100000-step MD diffusion and thermal conductivity values]. If the audited ranking is unchanged, we will retain a qualified interpretation that high-temperature training exposure can flatten the effective PES; if the ranking changes, the text and figure will be regenerated from the audited workflow. [Yi- I need you to explain to me what this last sentence means]

The current thermal conductivity curves are therefore treated as a protocol-audit target rather than a final transport conclusion. They still motivate the methodological point that validating MLFFs solely on static structures and short-time dynamics is insufficient: collective transport properties require matched simulation protocols, uncertainty estimates, and trajectory health checks before they can support model-ranking claims.

3.2 Non-Equilibrium Diffusion Pathways: The Critical Test of Generalization

The migration path benchmark evaluates a different measurement operator from the equilibrium RMSE. For each selected end-point pair $\tau = (a_s, a_t, c)$, the DFT NEB path defines a process-conditioned sequence of configurations $\Gamma^*_{\mathrm{D},\tau}$ that includes the metastable endpoints, intermediate distorted images, and the transition-state region. A candidate MLFF is then tested either by fixed-path scoring on this DFT geometry or by self-relaxed NEB optimization on its own PES, as defined in Sec. **??**. These two operators answer different questions: whether the model assigns correct energies to the reference path, and whether the model can stably discover a plausible path without being given the DFT images.

3.2.1 Local Migration Events

For the fixed-path component of the benchmark, we used the DFT-relaxed NEB images as a common geometric reference and evaluated each MLFF by single-point energies on those images. This corresponds to $\Delta E^{\ddagger}_{\mathrm{fixed}}(M, \tau)$ in Sec. **??**: the model is not allowed to move the path, so the resulting error isolates energetic fidelity along a DFT-supplied trajectory.

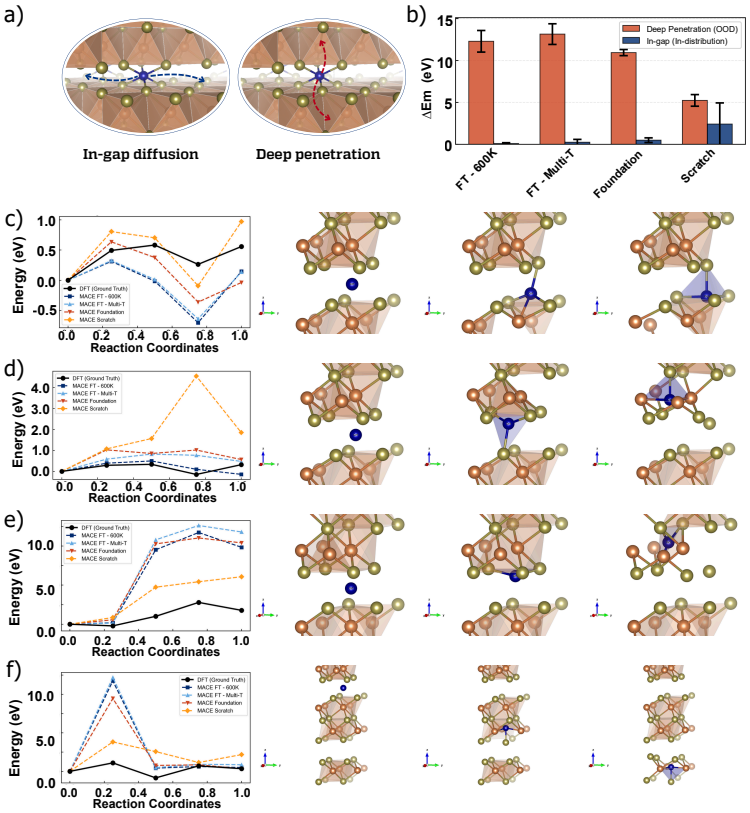


Fig. 3: **Comparative analysis of MLFF training strategies for predicting atomic migration barriers.** (a) Schematic illustration of the simulated Cr atom migration pathway between two stable sites within the Sb_2Te_3 bilayer. (b) Bar plot showing the migration energy prediction errors for various models, spanning bespoke training to advanced active- learning approaches. (c–f) Comparison of the minimum energy pathway (MEP) profiles for the Cr migration process. The solid black line denotes the ground-truth DFT reference data, while dashed lines represent predictions from MACE models trained with different strategies.

For the in-gap pathway, labeled 1–4 in the audited workflow, the current DFT candidate barrier is 0.336 eV. This value will be retained only after the restarted QE NEB run satisfies the stated convergence criterion; otherwise the table and plot will be regenerated from the converged restart. [Yi- Please explain what this sentence means too] The MACE Foundation fixed-path barrier is 1.024 eV, [Yi- do we have to quote these energy barriers to 3 d.p.? They aren't that accurate] giving an audited fixed-path error of +0.688 eV relative to the current DFT candidate. Thus, although the foundation model gives a smooth qualitative profile, it is not quantitatively accurate on this reference path.

The two baseline strategies illustrate different fixed-path failure modes. The MACE Scratch model strongly overestimates the barrier, consistent with insufficient sampling of high-energy, low-probability transition-state configurations in the training data. The MACE Foundation model produces a smoother qualitative profile but remains quantitatively miscalibrated on the DFT-supplied path, consistent with the possibility that broad pretraining captures generic PES smoothness while missing system-specific curvature along a rare-event pathway. [**?**] These results motivate fine-tuning, while the formal separation of fixed-path scoring and self-relaxed NEB optimization remains essential for interpreting the comparison. [Yi- please explain this sentence]

To probe the capability of generalizing the different MLFF training strategies, we evaluated their performance on two distinct Cr migration pathways (Fig. **??**a) with fundamentally different characteristics. The first pathway, *in-gap diffusion*, involves Cr migration within the vdW gap between Sb_2Te_3 quintuple layers, with low energy barriers and configurations similar to the training data, thereby testing interpolative accuracy. [Yi- more explanation please]

In contrast, the second pathway, *deep penetration*, requires the Cr atom to move vertically from the vdW gap and penetrate directly into the interior of a QL, ultimately reaching a deeply buried site within the layer. This migration path is associated with significantly higher energy barriers and accesses high-distortion configurations that are not well-represented in the training data, making it an out-of-distribution (OOD) scenario with potentially more metastable configurations along the pathway. Testing both pathways allows us to evaluate model accuracy on familiar configurations and extrapolative power in challenging regions of the configuration space.

In-Gap Diffusion: A Test of Interpolative Accuracy. For the in-gap diffusion pathway (Fig. **??**c–d), in which the Cr atom moves within the vdW gap or just shallowly penetrates the interface, the atomic environments along the minimum energy path (MEP) are expected to be closer to the near-equilibrium states sampled during the AIMD simulations than the deep-penetration path. Under the grouped-split, multi-seed re-evaluation (seeds 123/234/345), the absolute fixed-path barrier error on pathway 1–4 relative to the current DFT candidate (0.336 eV) is 0.474 ± 0.121 eV for FT–600K fine-tuned from the OMAT-FT checkpoint, 1.204 ± 0.270 eV for Scratch, and 0.678 ± 0.195 eV for Scratch-5%; a single-seed MACE-MPA-0 fine-tuned comparator gives 0.237 eV but is excluded from these seed-uncertainty statistics. FT–600K therefore remains the smallest-error fine-tuned model on this pathway, consistent with the single-model result reported previously, but the grouped, multi-seed re-evaluation does not reproduce the earlier point estimate of a 0.16 eV barrier error: the audited error under the present protocol is larger and has seed-to-seed spread. We treat the ranking among FT–600K, Scratch, and Scratch-5% on this single pathway as provisional evidence for task-relevant fine-tuning rather than a precise barrier-error claim, and we do not extend this pathway-1–4 comparison to the other audited pathways or to FT–MultiT, which was not retrained under the grouped protocol in this round.

To place the specialist models on the same fixed-path scale as current foundation checkpoints, we additionally evaluated ten single-checkpoint foundation models using the identical five-image pathway-1–4 inputs. The four representative checkpoints (MP-0b3, MPA0, OMAT-0 medium, and MH-1) predicted barriers of 1.503, 1.791, 1.602,

and 1.841 eV, respectively, compared with the provisional DFT value of 0.336050 eV. Thus, all four foundation representatives overestimate the barrier by 1.17–1.50 eV; MPA0 is not the kinetic-accuracy winner in this completed comparison. The individual profiles and the family-aggregated profiles are provided in the Supplementary Information (Fig. S**??**), while the full preflight, raw outputs, and fixed-path diagnostics are archived with the benchmark release. This comparison is deliberately reported as single-checkpoint evidence: foundation models do not provide seed-based uncertainty, and the declared overlap audit does not test unknown foundation-pretraining corpora.

Deep Penetration: A Test of Extrapolative Robustness. The "deep penetration" pathway provides a much more stringent test of model robustness. This path involves significant lattice distortion as the Cr atom pushes its way through the covalently bonded quintuple layer (QL), creating atomic environments that are far from those seen in the training data (Fig. **??**a).

The deep-penetration path samples a different region of μ_τ from the in-gap path: its transition-state images involve larger lattice distortion and weaker overlap with the AIMD training distribution. We therefore use it as a probe of extrapolative robustness rather than as a direct extension of the near-equilibrium RMSE test. Until the restarted QE references and multi-seed model rankings are closed, [Yi- needs explanation] the conservative conclusion is that near-equilibrium accuracy does not determine rare-event barrier accuracy.

The MACE Scratch model, which performed least well of the four models on the in-gap pathway, yields the lowest barrier error for this out-of-distribution task (Fig. **??**e, f). We do not interpret this isolated result as evidence that the scratch model is physically superior. Rather, because the deep-penetration path samples highly strained configurations and the current DFT NEB trajectories are still being consolidated, this comparison is treated as a question of seed- and path-stability. The completed analysis will report the multi-seed barrier distribution and the converged QE reference values for these paths [PLACEHOLDER: multiseed deep-penetration barrier errors and convergence status]. Until those checks are complete, the result supports the narrower conclusion that equilibrium accuracy does not reliably predict rare-event barrier accuracy.

The Critical Role of Task-Specific Fine-Tuning

The improvement seen in the fine-tuned models, especially for the in-gap diffusion task (Fig. **??**b), shows that adaptation can improve process fidelity when the fine-tuning data overlap with the task-conditioned pathway distribution. The MACE FT–600K model, fine-tuned on data generated at the target temperature of the migration event, gives the smallest reported in-gap barrier error in the current analysis. This supports the interpretation that task-relevant fine-tuning can recalibrate the local PES along a selected migration probe, while leaving open whether the same adaptation improves all non-equilibrium processes.

Within this case study, task relevance appears to be more predictive than data volume alone: the 600 K fine-tuned model is closer to the in-gap migration probe than the broader multi-temperature fine-tuned model. This interpretation depends on path-aware train–test proximity and the Scratch-5% control, which are included in the

acceptance criteria in Sec. **??**. The grouped-split, nearest-neighbor SOAP/RMSD audit described above finds no in-gap NEB-image overlap with either the full grouped training set or the Scratch-5% subset. The Scratch-5% control (same ∼1,000-configuration grouped FT-600K training subset, random initialization, seeds 123/234/345) gives a pathway-1–4 absolute barrier error of 0.678 ± 0.195 eV, intermediate between FT–600K-OMATFT (0.474 ± 0.121 eV) and full Scratch (1.204 ± 0.270 eV) on this single pathway; given the small sample (one pathway, three seeds, ∼90 training frames), we treat this as consistent with, rather than definitive proof of, a foundation-pretraining benefit beyond data volume alone. The corresponding comparison against FT–MultiT could not be updated in this round because FT–MultiT was not retrained under the grouped protocol.

The success of targeted fine-tuning points the way forward, while the divergent results of the 600 K and Multi-T models indicate that the data selection process is non-trivial. This motivates benchmark-guided data acquisition, including uncertainty-guided active learning, to identify configurations that improve model robustness and accuracy in a data-efficient manner.

3.2.2 NEB Stability as an Additional Robustness Criterion

We also evaluated the separate self-relaxed NEB operator $\Delta E^{\ddagger}_{\mathrm{self}}(M, \tau)$, in which the MLFF supplies the forces used to optimize the band. The 0.41 eV energy barrier value reported in Fig. S3 is the MACE Foundation model's self-consistent barrier on its own optimized path. It is, therefore, not a second DFT reference and not a contradiction of the +0.688 eV fixed-path error in Fig. **??**d. In contrast, the scratch, FT–600K, and FT–MultiT models exhibit force growth and early termination in this self-relaxed task, [Yi-we need to discuss this sentence. what is "force growth" and early termination?] making NEB stability itself an additional robustness criterion. Any low-barrier MLFF-relaxed path should be interpreted physically only after DFT single-point or DFT-restarted validation.

3.3 Interlayer Sliding: Probing Robustness to Non-Local, Collective Displacements

Having tested the models' response to local, high-distortion events, we evaluated their robustness to a different class of out-of-distribution challenge: large-scale, collective lattice displacements. To this end, we simulated the shearing of one Sb_2Te_3 layer relative to another, a process governed by the subtle corrugations of the interlayer vdW potential energy surface (Fig. **??**a). This task is particularly challenging for MLFFs as the local atomic environment of any given atom changes only minimally, while the global configuration undergoes a significant transformation that crosses periodic boundaries. The configurations along this sliding path were not present in the training data.

We first examined the case of pure, undoped Sb_2Te_3 (Fig. **??**b). The results reveal a clear trade-off between the models. The MACE Foundation model provides the most reasonable estimate of the energy barrier's shape and magnitude, suggesting its vast pre-training on bulk crystals has endowed it with a better implicit understanding of such periodic, mechanical deformations. However, it fails to maintain translational

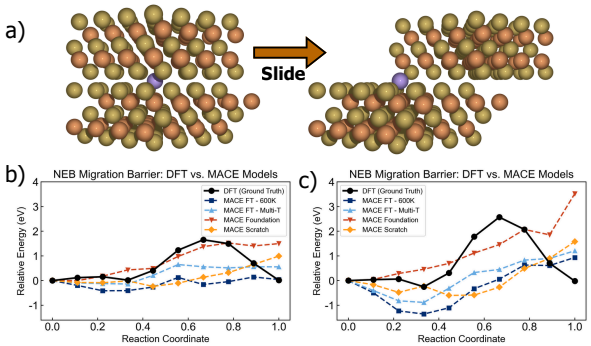


Fig. 4: Evaluation of MACE models on collective lattice displacement: interlayer sliding in Sb_2Te_3. (a) Schematic illustration of the bilayer sliding process in Sb_2Te_3, where the top layer (yellow/orange atoms) slides relative to the bottom layer along the crystallographic direction. The purple sphere indicates the position of a Cr dopant when present. (b) Energy barriers for interlayer sliding in pristine Sb_2Te_3 as calculated by DFT (black, ground truth) and various MACE models. The reaction coordinate represents the normalized sliding distance from the initial to final configuration. (c) Energy barriers for the same sliding process in Cr-doped Sb_2Te_3.

symmetry, incorrectly predicting the final state to be higher in energy than the identical initial state. This energy drift is a clear artifact, indicating a failure to perfectly respect the periodic nature of the simulation cell under large displacements. The MACE Scratch model exhibits a similar version of this artifact.

In contrast, the fine-tuned models (MACE FT–600K and Multi-T) significantly underestimate the energy barrier. This suggests a testable hypothesis: fine-tuning that improves local chemistry around the Cr dopant may reduce fidelity for weaker, longer-range interlayer physics inherited from the foundation model. The present data support this interpretation as a case-study observation rather than as a universal statement about fine-tuning.

This trend is also observed in the Cr-doped system (Fig. **??**c), suggesting that it is not solely an artifact of the pristine sliding setup. The failure of all models to perfectly capture both the barrier height and the end-point periodicity highlights a limitation relevant to local-descriptor-based MLFFs. Phenomena like shearing, stacking faults, and dislocation glide are inherently non-local. While the models can describe local bonding and coordination, they may struggle to enforce long-range physical constraints that extend beyond their cut-off radius. This further underscores the need for careful validation when using standard MLFFs to study mechanical properties and points towards future work in developing training sets that explicitly include these collective deformation modes or exploring architectures designed to capture long-range physics more effectively.

3.4 Representation Learning Analysis

To diagnose possible origins of the observed performance differences, we conducted a multi-faceted analysis of the learned representations. We projected high-dimensional atomic environment descriptors from each model's 600 K MD trajectory onto a two-dimensional space using two complementary visualization techniques: t-distributed stochastic neighbor-embedding (t-SNE) [**?**] to assess representational dissimilarity, and Potential of Heat-diffusion for Affinity-based Trajectory Embedding (PHATE) [**?**] to visualize trajectory geometry (Fig. **??**).

The t-SNE projection (Fig. **??**a) suggests that models trained with different strategies produce qualitatively distinct encodings. The MACE Foundation and MACE Scratch models occupy separated regions in the projection, while the two fine-tuned models occupy an intermediate region. Because low-dimensional embeddings can amplify or suppress apparent separation, we use these plots as qualitative diagnostics and report quantitative checks in the original descriptor space. In the audited analysis, the original 6191-dimensional Euclidean silhouette score is 0.333, while z-scoring the original features gives a silhouette score of -0.019; the projected baselines are 0.397 ± 0.027 for t-SNE and 0.473 for PHATE. These values support representational differences but they do not, by themselves, prove a unique mechanistic cause.

PHATE provides a complementary visualization of trajectory geometry, but we do not treat the projection as a direct mechanistic explanation. The PHATE projection (Fig. **??**b) maps the continuous time-evolution of the system to a low-dimensional trajectory manifold whose geometry is compared with the potential energy in Fig. **??**c. Along the primary axis of this manifold (PHATE 1), the specialist models (Scratch and FT) occupy similar regions, while differences along PHATE 2 separate the scratch-trained model from the fine-tuned models.

This separation is consistent with, but does not prove, a difference in how the model families represent configurations sampled during the dynamics at 600 K. The scratch-trained model's isolated region may reflect a narrower or less constrained representation, whereas the fine-tuned models' intermediate locations are consistent with adaptation from the pre-trained foundation model. Direct causal claims require perturbation or intervention tests beyond these projections.

This geometric difference in representation is consistent with the observed diffusion and NEB behavior, but we do not treat the embedding as a direct mechanistic proof. Diffusion is a rare-event process that requires the model to generalize to out-of-distribution transition states. In this context, the scratch model's separated representation is consistent with a less constrained PES outside its training domain, while the fine-tuned models' intermediate representations are consistent with adaptation of a general physical prior to the target system. We therefore interpret the latent-space analysis as a diagnostic for model families that should be read together with the task-level barrier, transport, and stability benchmarks.

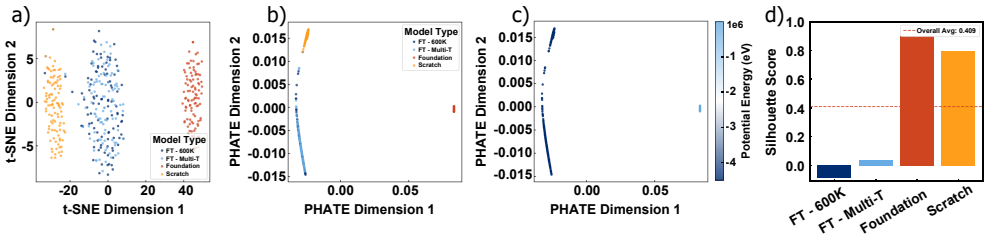


Fig. 5: **Representation diagnostics from trajectory embeddings and original-descriptor checks.** Projections of atomic environment descriptors from 600 K MD trajectories. **(a)** A t-SNE projection shows that Foundation (red) and Scratch (orange) models produce highly separated representations, while fine-tuned models (blue) are intermediate. **(b)** PHATE projection visualizes the system's continuous trajectory manifold. The separation between the scratch-model and fine-tuned-model projections along PHATE 2 is interpreted as a qualitative diagnostic rather than direct mechanistic proof. **(c)** The same PHATE embedding, colored by potential energy, compares projection geometry with model energy values. The foundation model's large energy offset indicates its nature as an uncalibrated prior. **(d)** Silhouette scores quantify separability in the original descriptor space and in low-dimensional embeddings. The projected values are used as visualization diagnostics, while the original-space values provide the primary quantitative check on representational dissimilarity.

3.5 Dissecting Learned Representations with Feature-Level Interpretability

While t-SNE and PHATE suggest that distinct training strategies produce separable descriptor patterns, these projections do not establish the physical principles learned by each model. To add a feature-level diagnostic, we use SHapley Additive exPlanations (SHAP) [**?**] on surrogate models of MACE's absolute energy error. This analysis identifies SOAP descriptors to which the surrogate error predictors are most sensitive; it does not directly explain the internal message-passing mechanism of MACE.

Our methodology is designed to identify descriptors correlated with predictive error. For each MACE model, we trained a Gradient-Boosting Regressor to predict absolute total-energy error from 390 SOAP and 17 structural descriptors extracted from the X-FORCE 150-frame trajectory; frame identity is joined explicitly through `image_index`. [**?**] This approach interprets a simpler surrogate of the error landscape, not the internal message-passing mechanism of MACE itself. The audited 5-fold cross-validation R^2 values are 0.9819, 0.8756, and 0.9730 for FT–600K, FT–Multi–T, and Scratch, respectively. We then applied TreeSHAP to these surrogate models, assigning feature-importance values to SOAP descriptors that are most correlated with prediction errors.

Our analysis shows a divergence in the surrogate feature-importance landscapes associated with each training strategy. A global overview of the top 50 influential descriptors across models (Fig. **??**b) shows that the fine-tuned multi-temperature (FT–Multi-T) surrogate concentrates importance more strongly, whereas the scratch-model

surrogate distributes importance more diffusely. In the locked revised ranking, the largest FT–Multi-T contribution is `Cr-Sb_n43_l0`; the leading Scratch contributions are dominated by Cr–Te descriptors. These are descriptive surrogate sensitivities, not evidence that the descriptors are causal or mechanistically internalized by MACE.

This disparity is quantified in Fig. **??**a, which compares the feature-importance profiles of the fine-tuned and scratch models. The FT–Multi-T feature is consistent with sensitivity to Cr–Sb local coordination in the doped lattice, but a direct physical assignment requires perturbation tests rather than SHAP rankings alone. The scratch model's broader surrogate-importance profile is consistent with a less concentrated error predictor; it does not by itself establish a physical explanation for the model's NEB behavior.

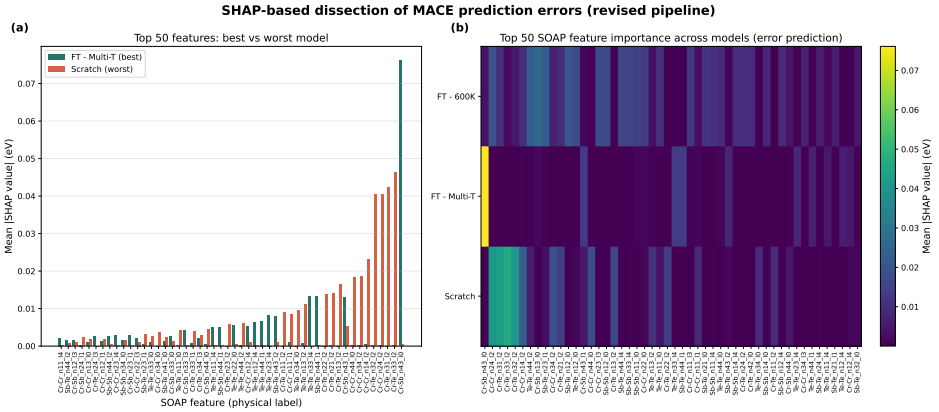


Fig. 6: **SHAP-based Dissection of Feature Importance in MACE Models.** **(a)** Mean absolute SHAP values for the top 50 SOAP features, comparing the best-performing fine-tuned model (FT–Multi-T) and the worst-performing scratch-trained model. The corrected audit identifies `Cr-Sb_n43_l0` as the largest mean absolute SHAP feature in the FT–Multi-T surrogate. **(b)** Heat map of surrogate feature importance for the top 50 SOAP features across all training strategies. The FT–Multi-T surrogate concentrates importance on fewer features, while the scratch-model surrogate distributes importance more diffusely.

4 Conclusions

We benchmarked specialist and foundation-based MLFFs for Cr-doped Sb_2Te_3 using the measurement-operator framework defined in Sec. **??**. This formalism separates equilibrium point-wise accuracy from task-conditioned process fidelity, and it distinguishes fixed-path energetic scoring from self-relaxed path optimization. Under this framework, diffusion trajectories and structured pathways show that models with similar equilibrium metrics can diverge sharply under kinetically relevant distortions.

Our findings support several broader hypotheses. Robust benchmarking should span both in-distribution and out-of-distribution tasks, as accuracy near equilibrium does not guarantee kinetic fidelity. Fine-tuning enhances targeted performance but can reduce global physical generality, emphasizing the need for carefully designed adaptation strategies. [Yi- can we make this sentence more easy to understand] Latent space analysis shows that different training paradigms can produce qualitatively distinct representations, helping to diagnose why models with similar traditional metrics may behave differently in downstream tasks. Notably, the deep penetration paths remain the most stringent OOD regime in this benchmark and are being finalized through restarted QE NEB calculations before final numerical ranking.

Together, these results indicate that adding "more data" is not necessarily sufficient; task-relevant, non-equilibrium sampling is often the decisive ingredient for meaningful improvement. Migration-inspired probes, coupled with latent space diagnostics and auditable data provenance, provide a practical route to identify which additional configurations are needed and where models are likely to fail. In this case study, the framework reframes the specialist–generalist debate: the goal is not to choose *between* model classes, but to exploit the complementary strengths they offer while validating them against the specific non-equilibrium tasks they are expected to perform.

Acknowledgments

- **Funding**
  This work was supported by the United States Department of Defense-funded Center of Excellence for Advanced Electro-photonics with 2D materials—Morgan State University, under Grant No. W911NF2120213. The authors thank Drs. Frank Gardea and Owen Vail, the cooperative agreement managers of the Center, for their interest and constructive comments. They also thank Dr. Ramesh Budhani for his guidance on the project. YC thanks Johns Hopkins University for their support in AY 24–25. Computational resource support was provided by the petascale Hopkins facility, Advanced Research Computing at Hopkins (ARCH) (rockfish.jhu.edu), supported by National Science Foundation award OAC 1920103. Partial funding for ARCH's infrastructure was originally provided by the State of Maryland.
- **Conflict of interest/Competing interests**
  The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.
- **Ethics approval and consent to participate**
  Not applicable. This study does not involve human participants or animal subjects.
- **Consent for publication**
  Not applicable.
- **Data availability**
  The supporting dataset, including training manifests, NEB path metadata, model-predicted barriers, and post-processing tables, will be released through MigrationBench on Hugging Face at https://huggingface.co/datasets/alinacao2000/MigrationBench [PLACEHOLDER: final dataset DOI or commit hash]. The manifest records the source trajectory, split assignment, random seed, model provenance,

DFT/MLFF task type, convergence status, and checksum for each derived artifact. NEB paths are evaluation/reference data and were not merged into the fine-tuning datasets. The frame-level force audit covers 9,163 released training frames and found no unclassified or NEB-derived all-zero-force frame; the nine exact zero-force records are explicitly labelled isolated-atom E_0 references (release commit `f5c6d19`).

- **Materials availability**
  Not applicable. This work is entirely computational and does not involve the synthesis or distribution of physical materials.
- **Code availability**
  The scripts used for data processing, DFT/MD post-processing, and plotting are available on GitHub https://github.com/yicao-elina/MigrationBench.git. The Sec. 3.5/Fig. 6 SHAP chain is released as `scripts/shap_analysis.py`, `scripts/shap_feature_labels.py`, and `scripts/plot_shap_fig6.py`, using the compact three-model X-FORCE inputs in `data/raw_compact/revision1/shap/`. The source scope and MD5s are recorded in `docs/audit/shap_source_lineage.md`; the four-model, 2-concentration, and Dual-X ID/OOD/NEB workflows are not inputs to Fig. 6.
- **Author contribution**
  Y. C.: conceptualization, methodology, data curation, DFT and MD simulations, analysis, visualization, and manuscript-writing. **P. M.**: literature review, conceptualization, and manuscript- writing. **P. C.**: conceptualization, project administration, supervision, funding acquisition, and manuscript-writing.

References

[] Unke, O.T., Chmiela, S., Sauceda, H.E., Gastegger, M., Poltavsky, I., Schütt, K.T., Tkatchenko, A., Müller, K.-R.: Machine Learning Force Fields. Chemical Reviews **121**(16), 10142–10186 (2021) https://doi.org/10.1021/acs.chemrev.0c01111 . Publisher: American Chemical Society. Accessed 2025-05-15

[] Smith, J.S., Isayev, O., Roitberg, A.E.: Ani-1: an extensible neural network potential with dft accuracy at force field computational cost. Chemical science **8**(4), 3192–3203 (2017)

[] Chen, C., Deng, Z., Tran, R., Tang, H., Chu, I.-H., Ong, S.P.: Accurate force field for molybdenum by machine learning large materials data. Physical Review Materials **1**(4), 043603 (2017) https://doi.org/10.1103/PhysRevMaterials.1.043603 . Publisher: American Physical Society. Accessed 2025-08-20

[3] Batatia, I., Kovacs, D.P., Simm, G., Ortner, C., Csányi, G.: Mace: Higher order equivariant message passing neural networks for fast and accurate force fields. Advances in neural information processing systems **35**, 11423–11436 (2022)

[] Vandermause, J., Xie, Y., Lim, J.S., Owen, C.J., Kozinsky, B.: Active learning of reactive Bayesian force fields applied to heterogeneous catalysis dynamics of H/Pt. Nature Communications **13**(1), 5183 (2022) https://doi.org/10.1038/

s41467-022-32294-0 . Accessed 2024-09-29

[] Batzner, S., Musaelian, A., Sun, L., Geiger, M., Mailoa, J.P., Kornbluth, M., Molinari, N., Smidt, T.E., Kozinsky, B.: E(3)-equivariant graph neural networks for data-efficient and accurate interatomic potentials. Nature Communications **13**(1), 2453 (2022) https://doi.org/10.1038/s41467-022-29939-5 . Publisher: Nature Publishing Group. Accessed 2024-12-22

[] Bartók, A.P., Payne, M.C., Kondor, R., Csányi, G.: Gaussian Approximation Potentials: The Accuracy of Quantum Mechanics, without the Electrons. Physical Review Letters **104**(13), 136403 (2010) https://doi.org/10.1103/PhysRevLett.104.136403 . Publisher: American Physical Society. Accessed 2025-08-20

[] Friederich, P., Häse, F., Proppe, J., Aspuru-Guzik, A.: Machine-learned potentials for next-generation matter simulations. Nature Materials **20**(6), 750–761 (2021)

[] Zeng, J., Zhang, D., Lu, D., Mo, P., Li, Z., Chen, Y., Rynik, M., Huang, L., Li, Z., Shi, S., Wang, Y., Ye, H., Tuo, P., Yang, J., Ding, Y., Li, Y., Tisi, D., Zeng, Q., Bao, H., Xia, Y., Huang, J., Muraoka, K., Wang, Y., Chang, J., Yuan, F., Bore, S.L., Cai, C., Lin, Y., Wang, B., Xu, J., Zhu, J.-X., Luo, C., Zhang, Y., Goodall, R.E.A., Liang, W., Singh, A.K., Yao, S., Zhang, J., Wentzcovitch, R., Han, J., Liu, J., Jia, W., York, D.M., E, W., Car, R., Zhang, L., Wang, H.: DeePMD-kit v2: A software package for deep potential models. The Journal of Chemical Physics **159**(5), 054801 (2023) https://doi.org/10.1063/5.0155600 . Accessed 2024-12-15

[] Wines, D., Choudhary, K.: CHIPS-FF: Evaluating Universal Machine Learning Force Fields for Material Properties. ACS Materials Letters, 2105–2114 (2025) https://doi.org/10.1021/acsmaterialslett.5c00093 . Publisher: American Chemical Society. Accessed 2025-05-11

[] Musaelian, A., Batzner, S., Johansson, A., Sun, L., Owen, C.J., Kornbluth, M., Kozinsky, B.: Learning local equivariant representations for large-scale atomistic dynamics. Nature Communications **14**(1), 579 (2023) https://doi.org/10.1038/s41467-023-36329-y . Publisher: Nature Publishing Group. Accessed 2025-09-09

[] Chen, C., Ong, S.P.: A universal graph deep learning interatomic potential for the periodic table. Nature Computational Science **2**(11), 718–728 (2022) https://doi.org/10.1038/s43588-022-00349-3 . Publisher: Springer Science and Business Media LLC. Accessed 2025-07-27

[] Lindsey, R.K., Fried, L.E., Goldman, N.: Chimes: A force matched potential with explicit three-body interactions for molten carbon. Journal of chemical theory and computation **13**(12), 6222–6229 (2017)

[] Xie, T., Grossman, J.C.: Crystal graph convolutional neural networks for an accurate and interpretable prediction of material properties. Physical review letters **120**(14), 145301 (2018)

[] Deng, B., Zhong, P., Jun, K., Riebesell, J., Han, K., Bartel, C.J., Ceder, G.: CHGNet as a pretrained universal neural network potential for charge-informed atomistic modelling. Nature Machine Intelligence **5**(9), 1031–1041 (2023) https://doi.org/10.1038/s42256-023-00716-3 . Publisher: Nature Publishing Group. Accessed 2025-08-01

[] Chen, C., Ye, W., Zuo, Y., Zheng, C., Ong, S.P.: Graph Networks as a Universal Machine Learning Framework for Molecules and Crystals. Chemistry of Materials **31**(9), 3564–3572 (2019) https://doi.org/10.1021/acs.chemmater.9b01294 . Publisher: American Chemical Society. Accessed 2025-07-09

[] Yang, H., Hu, C., Zhou, Y., Liu, X., Shi, Y., Li, J., Li, G., Chen, Z., Chen, S., Zeni, C., Horton, M., Pinsler, R., Fowler, A., Zügner, D., Xie, T., Smith, J., Sun, L., Wang, Q., Kong, L., Liu, C., Hao, H., Lu, Z.: MatterSim: A Deep Learning Atomistic Model Across Elements, Temperatures and Pressures. arXiv. arXiv:2405.04967 [cond-mat] (2024). https://doi.org/10.48550/arXiv.2405.04967 . http://arxiv.org/abs/2405.04967 Accessed 2025-04-30

[] Wood, B.M., Dzamba, M., Fu, X., Gao, M., Shuaibi, M., Barroso-Luque, L., Abdelmaqsoud, K., Gharakhanyan, V., Kitchin, J.R., Levine, D.S., Michel, K., Sriram, A., Cohen, T., Das, A., Rizvi, A., Sahoo, S.J., Ulissi, Z.W., Zitnick, C.L.: UMA: A Family of Universal Models for Atoms. arXiv. arXiv:2506.23971 [cs] (2025). https://doi.org/10.48550/arXiv.2506.23971 . http://arxiv.org/abs/2506.23971 Accessed 2025-08-20

[] Merchant, A., Batzner, S., Schoenholz, S.S., Aykol, M., Cheon, G., Cubuk, E.D.: Scaling deep learning for materials discovery. Nature **624**(7990), 80–85 (2023) https://doi.org/10.1038/s41586-023-06735-9 . Publisher: Nature Publishing Group. Accessed 2025-09-09

[] Zhang, L.: Deep Potential Molecular Dynamics: A Scalable Model with the Accuracy of Quantum Mechanics. Physical Review Letters **120**(14) (2018) https://doi.org/10.1103/PhysRevLett.120.143001

[] Smith, J.S., Nebgen, B.T., Zubatyuk, R., Lubbers, N., Devereux, C., Barros, K., Tretiak, S., Isayev, O., Roitberg, A.E.: Approaching coupled cluster accuracy with a general-purpose neural network potential through transfer learning. Nature communications **10**(1), 2903 (2019)

[] Chmiela, S., Sauceda, H.E., Poltavsky, I., Müller, K.-R., Tkatchenko, A.: sGDML: Constructing accurate and data efficient molecular force fields using machine learning. Computer Physics Communications **240**, 38–45 (2019) https://doi.org/10.1016/j.cpc.2019.02.007

[] Dunn, A., Wang, Q., Ganose, A., Dopp, D., Jain, A.: Benchmarking materials property prediction methods: the Matbench test set and Automatminer reference algorithm. npj Computational Materials **6**(1), 138 (2020) https://doi.org/10.1038/

s41524-020-00406-3 . Publisher: Nature Publishing Group. Accessed 2025-08-12

[] Choudhary, K., Wines, D., Li, K., Garrity, K.F., Gupta, V., Romero, A.H., Krogel, J.T., Saritas, K., Fuhr, A., Ganesh, P., *et al.*: Jarvis-leaderboard: a large scale benchmark of materials design methods. npj Computational Materials **10**(1), 93 (2024)

[] Betala, S., Gleason, S.P., Ramlaoui, A., Xu, A., Channing, G., Levy, D., Fourrier, C., Kazeev, N., Joshi, C.K., Kaba, S.-O., et al.: Lemat-genbench: A unified evaluation framework for crystal generative models. arXiv preprint arXiv:2512.04562 (2025)

[] Friederich, P., Häse, F., Proppe, J., Aspuru-Guzik, A.: Machine-learned potentials for next-generation matter simulations. Nature Materials **20**(6), 750–761 (2021) https://doi.org/10.1038/s41563-020-0777-6

[] Deringer, V.L., Caro, M.A., Csányi, G.: Machine Learning Interatomic Potentials as Emerging Tools for Materials Science. Advanced Materials **31**(46), 1902765 (2019) https://doi.org/10.1002/adma.201902765

[] Jacobs, R., Morgan, D., Attarian, S., Meng, J., Shen, C., Wu, Z., Xie, C.Y., Yang, J.H., Artrith, N., Blaiszik, B., *et al.*: A practical guide to machine learning interatomic potentials–status and future. Current Opinion in Solid State and Materials Science **35**, 101214 (2025)

[] Schran, C., Thiemann, F.L., Rowe, P., Müller, E.A., Marsalek, O., Michaelides, A.: Machine learning potentials for complex aqueous systems made simple. Proceedings of the National Academy of Sciences **118**(38), 2110077118 (2021) https://doi.org/10.1073/pnas.2110077118

[] Kapil, V., Schran, C., Zen, A., Chen, J., Pickard, C.J., Michaelides, A.: The first-principles phase diagram of monolayer nanoconfined water. Nature **609**(7927), 512–516 (2022) https://doi.org/10.1038/s41586-022-05036-x

[] Grisafi, A., Fabrizio, A., Meyer, B., Wilkins, D.M., Corminboeuf, C., Ceriotti, M.: Transferable Machine-Learning Model of the Electron Density. ACS Central Science **5**(1), 57–64 (2019) https://doi.org/10.1021/acscentsci.8b00551

[] Liu, Y., He, X., Mo, Y.: Discrepancies and error evaluation metrics for machine learning interatomic potentials. npj Computational Materials **9**(1), 174 (2023)

[] Janet, J.P., Liu, F., Nandy, A., Duan, C., Yang, T., Lin, S., Kulik, H.J.: Designing in the Face of Uncertainty: Exploiting Electronic Structure and Machine Learning Models for Discovery in Inorganic Chemistry. Inorganic Chemistry **58**(16), 10592–10606 (2019) https://doi.org/10.1021/acs.inorgchem.9b00109

[] Lookman, T., Balachandran, P.V., Xue, D., Yuan, R.: Active learning in materials

science with emphasis on adaptive sampling using uncertainties for targeted design. npj Computational Materials **5**(1), 21 (2019) https://doi.org/10.1038/s41524-019-0153-8

[] Zuo, Y., Chen, C., Li, X., Deng, Z., Chen, Y., Behler, J., Csányi, G., Shapeev, A.V., Thompson, A.P., Wood, M.A., Ong, S.P.: Performance and Cost Assessment of Machine Learning Interatomic Potentials. The Journal of Physical Chemistry A **124**(4), 731–745 (2020) https://doi.org/10.1021/acs.jpca.9b08723

[] Jinnouchi, R., Lahnsteiner, J., Karsai, F., Kresse, G., Bokdam, M.: Phase Transitions of Hybrid Perovskites Simulated by Machine-Learning Force Fields Trained on the Fly with Bayesian Inference. Physical Review Letters **122**(22), 225701 (2019) https://doi.org/10.1103/PhysRevLett.122.225701

[] Zeni, C., Rossi, K., Pavloudis, T., Kioseoglou, J., de Gironcoli, S., Palmer, R.E., Baletto, F.: Data-driven simulation and characterisation of gold nanoparticle melting. Nature Communications **12**(1), 6056 (2021) https://doi.org/10.1038/s41467-021-26199-7

[] Vandermause, J., Torrisi, S.B., Batzner, S., Xie, Y., Sun, L., Kolpak, A.M., Kozinsky, B.: On-the-fly active learning of interpretable Bayesian force fields for atomistic rare events. npj Computational Materials **6**(1), 20 (2020) https://doi.org/10.1038/s41524-020-0283-z

[] Bernstein, N., Bhattarai, B., Csányi, G., Drabold, D.A., Elliott, S.R., Deringer, V.L.: Quantifying Chemical Structure and Machine-Learned Atomic Energies in Amorphous and Liquid Silicon. Angewandte Chemie **131**(21), 7131–7135 (2019) https://doi.org/10.1002/ange.201902625

[] Zhang, L., Lin, D.-Y., Wang, H., Car, R., E, W.: Active learning of uniformly accurate interatomic potentials for materials simulation. Physical Review Materials **3**(2), 023804 (2019) https://doi.org/10.1103/PhysRevMaterials.3.023804

[] He, X., Zhu, Y., Epstein, A., Mo, Y.: Statistical variances of diffusional properties from ab initio molecular dynamics simulations. npj Computational Materials **4**(1), 18 (2018)

[] Fu, X., Wu, Z., Wang, W., Xie, T., Keten, S., Gomez-Bombarelli, R., Jaakkola, T.: Forces are not enough: Benchmark and critical evaluation for machine learning force fields with molecular simulations. arXiv preprint arXiv:2210.07237 (2022)

[] Mannan, S., Gonzales, C., Bihani, V., Lee, K.L.K., Gosvami, N.N., Miret, S., Krishnan, N.A.: Evaluating universal interatomic potentials for molecular dynamics of real-world minerals. In: AI for Accelerated Materials Design-ICLR 2025 (2025)

[] Miskin, K., Cao, Y., Marland, M., Shaikh, F., Moore, D.T., Marohn, J.A., Clancy, P.: Low-energy pathways lead to self-healing defects in cspbbr 3. Physical

Chemistry Chemical Physics **27**(29), 15446–15459 (2025)

[] Ruttinger, A.W., Sharma, D., Clancy, P.: Protocol for Directing Nudged Elastic Band Calculations to the Minimum Energy Pathway: Nurturing Errant Calculations Back to Convergence. Journal of Chemical Theory and Computation **18**(5), 2993–3005 (2022) https://doi.org/10.1021/acs.jctc.1c00926 . Publisher: American Chemical Society. Accessed 2024-08-30

[] Cao, Y., Clancy, P.: Migration as a probe: A generalizable benchmark framework for specialist vs. generalist machine-learned force fields in doped materials. arXiv preprint arXiv:2509.00090 (2025)

[] He, X., Zhu, Y., Mo, Y.: Origin of fast ion diffusion in super-ionic conductors. Nature communications **8**(1), 15893 (2017)

[] Bheemaguli, A.K., Xiao, P., Sai Gautam, G.: Evaluation of foundational machine learned interatomic potentials for migration barrier predictions. Digital Discovery **5**, 1809–1819 (2026) https://doi.org/10.1039/D5DD00534E

[] Novoselov, K.S., Mishchenko, A., Carvalho, A., Castro Neto, A.H.: 2D materials and van der Waals heterostructures. Science **353**(6298), 9439 (2016) https://doi.org/10.1126/science.aac9439 . Publisher: American Association for the Advancement of Science. Accessed 2025-09-09

[] Manzeli, S., Ovchinnikov, D., Pasquier, D., Yazyev, O.V., Kis, A.: 2D transition metal dichalcogenides. Nature Reviews Materials **2**(8), 1–15 (2017) https://doi.org/10.1038/natrevmats.2017.33 . Publisher: Nature Publishing Group. Accessed 2025-05-14

[] Akinwande, D., Brennan, C.J., Bunch, J.S., Egberts, P., Felts, J.R., Gao, H., Huang, R., Kim, J.-S., Li, T., Li, Y., *et al.*: A review on mechanics and mechanical properties of 2d materials—graphene and beyond. Extreme Mechanics Letters **13**, 42–77 (2017)

[] Guo, H.-W., Hu, Z., Liu, Z.-B., Tian, J.-G.: Stacking of 2D Materials. Advanced Functional Materials **31**(4), 2007810 (2021) https://doi.org/10.1002/adfm.202007810 . _eprint: https://advanced.onlinelibrary.wiley.com/doi/pdf/10.1002/adfm.202007810. Accessed 2025-08-30

[] Liu, A., Zhang, X., Liu, Z., Li, Y., Peng, X., Li, X., Qin, Y., Hu, C., Qiu, Y., Jiang, H., Wang, Y., Li, Y., Tang, J., Liu, J., Guo, H., Deng, T., Peng, S., Tian, H., Ren, T.-L.: The Roadmap of 2D Materials and Devices Toward Chips. Nano-Micro Letters **16**(1), 1–96 (2024) https://doi.org/10.1007/s40820-023-01273-5 . Publisher: Springer. Accessed 2025-08-30

[] He, Q.L., Hughes, T.L., Armitage, N.P., Tokura, Y., Wang, K.L.: Topological spintronics and magnetoelectronics. Nature Materials **21**(1), 15–23 (2022) https://doi.org/10.1038/s41563-021-01138-5 . Publisher: Nature Publishing Group. Accessed 2025-09-09

[] Qian, Q.: Van der Waals integration: Enables quantum explorations and innovative devices. MRS Bulletin **49**(4), 385–390 (2024) https://doi.org/10.1557/s43577-024-00668-y . Number: 4 Publisher: Springer. Accessed 2025-07-30

[] Song, J.C.W., Gabor, N.M.: Electron quantum metamaterials in van der Waals heterostructures. Nature Nanotechnology **13**(11), 986–993 (2018) https://doi.org/10.1038/s41565-018-0294-9 . Publisher: Nature Publishing Group. Accessed 2025-07-30

[] Nayak, C.: Microsoft unveils Majorana 1, the world's first quantum processor powered by topological qubits (2025). https://azure.microsoft.com/en-us/blog/quantum/2025/02/19/microsoft-unveils-majorana-1-the-worlds-first-quantum-processor-powered-by-topological-qubits/ Accessed 2025-04-08

[] Jin, Q., Jiang, S., Zhao, Y., Wang, D., Qiu, J., Tang, D.-M., Tan, J., Sun, D.-M., Hou, P.-X., Chen, X.-Q., Tai, K., Gao, N., Liu, C., Cheng, H.-M., Jiang, X.: Flexible layer-structured Bi2Te3 thermoelectric on a carbon nanotube scaffold. Nature Materials **18**(1), 62–68 (2019) https://doi.org/10.1038/s41563-018-0217-z . Publisher: Nature Publishing Group. Accessed 2025-07-22

[] Gautam, R., Hirai, T., Alasli, A., Nagano, H., Ohkubo, T., Uchida, K.-i., Sepehri-Amin, H.: Creation of flexible spin-caloritronic material with giant transverse thermoelectric conversion by nanostructure engineering. Nature Communications **15**(1), 2184 (2024) https://doi.org/10.1038/s41467-024-46475-6 . Publisher: Nature Publishing Group. Accessed 2025-04-23

[] Whittingham, M.S.: Lithium Batteries and Cathode Materials. Chemical Reviews **104**(10), 4271–4302 (2004) https://doi.org/10.1021/cr020731c . Publisher: American Chemical Society. Accessed 2025-09-09

[] Winter, M., Besenhard, J.O., Spahr, M.E., Novák, P.: Insertion Electrode Materials for Rechargeable Lithium Batteries. Advanced Materials **10**(10), 725–763 (1998) https://doi.org/10.1002/(SICI)1521-4095(199807)10:10⟨725::AID-ADMA725⟩3.0.CO;2-Z . _eprint: https://advanced.onlinelibrary.wiley.com/doi/pdf/10.1002/%28SICI%291521-4095%28199807%2910%3A10%3C725%3A%3AAID-ADMA725%3E3.0.CO%3B2-Z. Accessed 2025-09-09

[] Hu, W., Shen, J., Wang, T., Li, Z., Xu, Z., Lou, Z., Qi, H., Yan, J., Wang, J., Le, T., Zheng, X., Lu, Y., Lin, X.: Lithium Ion Intercalation-Induced Metal-Insulator Transition in Inclined-Standing Grown 2D Non-Layered Cr2S3 Nanosheets. Small

Methods **8**(12), 2400312 (2024) https://doi.org/10.1002/smtd.202400312 . _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1002/smtd.202400312. Accessed 2025-05-18

[] Gibertini, M., Koperski, M., Morpurgo, A.F., Novoselov, K.S.: Magnetic 2D materials and heterostructures. Nature Nanotechnology **14**(5), 408–419 (2019) https://doi.org/10.1038/s41565-019-0438-6 . Publisher: Nature Publishing Group. Accessed 2025-09-09

[] Tokura, Y., Yasuda, K., Tsukazaki, A.: Magnetic topological insulators. Nature Reviews Physics **1**(2), 126–143 (2019) https://doi.org/10.1038/s42254-018-0011-5 . Publisher: Nature Publishing Group. Accessed 2025-09-09

[] Zhang, H., Liu, C.-X., Qi, X.-L., Dai, X., Fang, Z., Zhang, S.-C.: Topological insulators in Bi2Se3, Bi2Te3 and Sb2Te3 with a single Dirac cone on the surface. Nature Physics **5**(6), 438–442 (2009) https://doi.org/10.1038/nphys1270 . Publisher: Nature Publishing Group. Accessed 2025-02-12

[] Guo, Y., Liu, Z., Peng, H.: A Roadmap for Controlled Production of Topological Insulator Nanostructures and Thin Films. Small **11**(27), 3290–3305 (2015) https://doi.org/10.1002/smll.201403426 . _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1002/smll.201403426. Accessed 2025-05-15

[] Checkelsky, J.G., Yoshimi, R., Tsukazaki, A., Takahashi, K.S., Kozuka, Y., Falson, J., Kawasaki, M., Tokura, Y.: Trajectory of the anomalous Hall effect towards the quantized state in a ferromagnetic topological insulator. Nature Physics **10**(10), 731–736 (2014) https://doi.org/10.1038/nphys3053 . Publisher: Nature Publishing Group. Accessed 2025-09-09

[] Kulbachinskii, V.A., Kytin, V.G., Kudryashov, A.A., Tarasov, P.M.: Thermoelectric properties of Bi2Te3, Sb2Te3 and Bi2Se3 single crystals with magnetic impurities. Journal of Solid State Chemistry **193**, 47–52 (2012) https://doi.org/10.1016/j.jssc.2012.03.042 . Accessed 2025-04-11

[] Haazen, P.P.J., Laloë, J.-B., Nummy, T.J., Swagten, H.J.M., Jarillo-Herrero, P., Heiman, D., Moodera, J.S.: Ferromagnetism in thin-film Cr-doped topological insulator Bi2Se3. Applied Physics Letters **100**(8), 082404 (2012) https://doi.org/10.1063/1.3688043 . Accessed 2025-03-17

[] Klimovskikh, I.I., Otrokov, M.M., Estyunin, D., Eremeev, S.V., Filnov, S.O., Koroleva, A., Shevchenko, E., Voroshnin, V., Rybkin, A.G., Rusinov, I.P., Blanco-Rey, M., Hoffmann, M., Aliev, Z.S., Babanly, M.B., Amiraslanov, I.R., Abdullayev, N.A., Zverev, V.N., Kimura, A., Tereshchenko, O.E., Kokh, K.A., Petaccia, L., Di Santo, G., Ernst, A., Echenique, P.M., Mamedov, N.T., Shikin, A.M., Chulkov, E.V.: Tunable 3D/2D magnetism in the (MnBi2Te4)(Bi2Te3)m topological insulators family. npj Quantum Materials **5**(1), 54 (2020) https://doi.org/10.1038/

s41535-020-00255-9 . Publisher: Nature Publishing Group. Accessed 2025-09-09

[] Fu, J., Huang, J., Bernard, F.: Electronic structure, elastic and optical properties of Bi2Te3/Sb2Te3 thermoelectric composites in the periodic-superlattice thin films. Composites Communications **28**, 100917 (2021) https://doi.org/10.1016/j.coco.2021.100917 . Accessed 2025-06-03

[] Sun, Y., Qin, H., Zhang, C., Wu, H., Yin, L., Liu, Z., Guo, S., Zhang, Q., Cai, W., Wu, H., Guo, F., Sui, J.: Sb2Te3 based alloy with high thermoelectric and mechanical performance for low-temperature energy harvesting. Nano Energy **107**, 108176 (2023) https://doi.org/10.1016/j.nanoen.2023.108176 . Accessed 2024-07-08

[] Zhang, L., Helm, T., Lin, H., Fan, F., Le, C., Sun, Y., Markou, A., Felser, C.: Quantum Oscillations in Ferromagnetic (Sb, V)2Te3 Topological Insulator Thin Films. Advanced Materials **33**(41), 2102107 (2021) https://doi.org/10.1002/adma.202102107 . _eprint: https://advanced.onlinelibrary.wiley.com/doi/pdf/10.1002/adma.202102107. Accessed 2025-09-09

[] Cortie, D., Zhao, W., Yue, Z., Li, Z., Bake, A., Marenych, O., Pastuovic, Z., Nancarrow, M., Zhang, Z., Qi, D.-C., Evans, P., Mitchell, D.R.G., Wang, X.: Creating thin magnetic layers at the surface of Sb2Te3 topological insulators using a low-energy chromium ion beam. Applied Physics Letters **116**(19), 192410 (2020) https://doi.org/10.1063/5.0006447 . Accessed 2024-11-13

[] Han, M.-K., Ryu, H., Kim, S.-J.: Effect of Chromium Doping on the Thermoelectric Properties of Bi2Te3: CrxBi2Te3 and CrxBi2-xTe3. Journal of Electronic Materials **42**(9), 2758–2763 (2013) https://doi.org/10.1007/s11664-013-2670-4 . Accessed 2025-03-16

[] A. Deus, D.P., Oliveira, I.S.S., Oliveira, J.B., Scopel, W.L., Miwa, R.H.: Magnetic switch and electronic properties in chromium-intercalated two-dimensional ${\mathrm{GeP}}_{3}$. Physical Review Materials **5**(5), 054002 (2021) https://doi.org/10.1103/PhysRevMaterials.5.054002 . Publisher: American Physical Society. Accessed 2025-07-24

[] Zhang, P., Liao, W., Zhu, Z., Qin, M., Zhang, Z., Jin, D., Liu, Y., Wang, Z., Lu, Z., Xiong, R.: Tuning the lattice thermal conductivity of Sb2Te3 by Cr doping: a deep potential molecular dynamics study. Physical Chemistry Chemical Physics (2023) https://doi.org/10.1039/d3cp00999h

[] Thompson, A.P., Aktulga, H.M., Berger, R., Bolintineanu, D.S., Brown, W.M., Crozier, P.S., Veld, P.J., Kohlmeyer, A., Moore, S.G., Nguyen, T.D., Shan, R., Stevens, M.J., Tranchida, J., Trott, C., Plimpton, S.J.: LAMMPS - a flexible simulation tool for particle-based materials modeling at the atomic, meso, and continuum scales. Computer Physics Communications **271**, 108171 (2022) https:

//doi.org/10.1016/j.cpc.2021.108171 . Accessed 2024-07-18

[] Kohn, W., Sham, L.J.: Self-Consistent Equations Including Exchange and Correlation Effects. Physical Review **140**(4A), 1133–1138 (1965) https://doi.org/10.1103/PhysRev.140.A1133 . Publisher: American Physical Society. Accessed 2025-08-20

[] Perdew, J.P., Burke, K., Ernzerhof, M.: Generalized Gradient Approximation Made Simple. Physical Review Letters **77**(18), 3865–3868 (1996) https://doi.org/10.1103/PhysRevLett.77.3865 . Publisher: American Physical Society. Accessed 2024-11-01

[] Sohier, T., Calandra, M., Mauri, F.: Density functional perturbation theory for gated two-dimensional heterostructures: Theoretical developments and application to flexural phonons in graphene. Physical Review B **96**(7), 075448 (2017) https://doi.org/10.1103/PhysRevB.96.075448 . Publisher: American Physical Society. Accessed 2025-03-19

[] Choudhary, K., Kalish, I., Beams, R., Tavazza, F.: High-throughput Identification and Characterization of Two-dimensional Materials using Density functional theory. Scientific Reports **7**(1), 5179 (2017) https://doi.org/10.1038/s41598-017-05402-0 . Publisher: Nature Publishing Group. Accessed 2024-11-19

[] Jain, A., Ong, S.P., Hautier, G., Chen, W., Richards, W.D., Dacek, S., Cholia, S., Gunter, D., Skinner, D., Ceder, G., et al.: Commentary: The materials project: A materials genome approach to accelerating materials innovation. APL materials **1**(1) (2013)

[] Deng, B., Choi, Y., Zhong, P., Riebesell, J., Anand, S., Li, Z., Jun, K., Persson, K.A., Ceder, G.: Systematic softening in universal machine learning interatomic potentials. npj Computational Materials **11**(1), 9 (2025) https://doi.org/10.1038/s41524-024-01500-6

[] Maaten, L.v.d., Hinton, G.: Visualizing data using t-sne. Journal of machine learning research **9**(Nov), 2579–2605 (2008)

[] Moon, K.R., Van Dijk, D., Wang, Z., Gigante, S., Burkhardt, D.B., Chen, W.S., Yim, K., Elzen, A.v.d., Hirn, M.J., Coifman, R.R., *et al.*: Visualizing structure and transitions in high-dimensional biological data. Nature biotechnology **37**(12), 1482–1492 (2019)

[] Lundberg, S., Lee, S.-I.: A Unified Approach to Interpreting Model Predictions. arXiv. arXiv:1705.07874 [cs] (2017). https://doi.org/10.48550/arXiv.1705.07874 . http://arxiv.org/abs/1705.07874 Accessed 2025-09-30

[] Bartók, A.P., Kondor, R., Csányi, G.: On representing chemical environments. arXiv preprint arXiv:1209.3140 (2012)